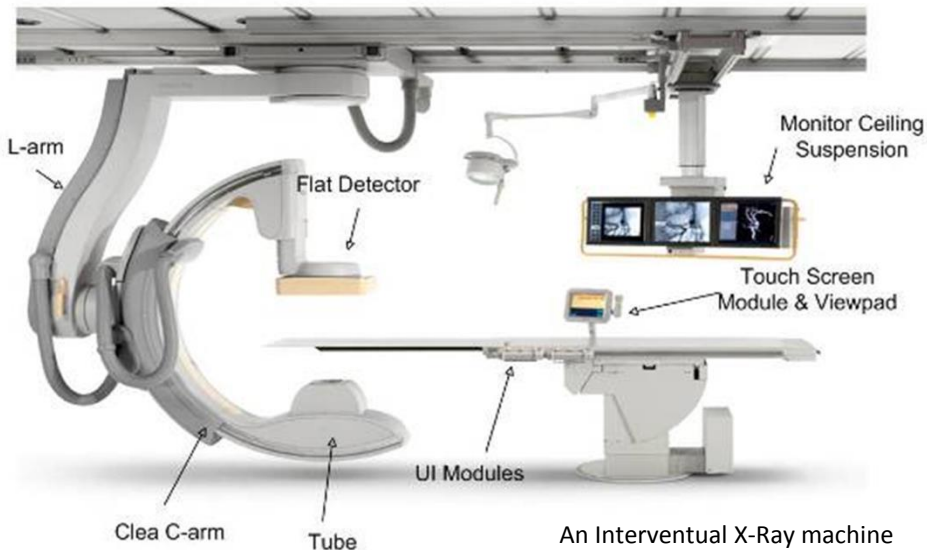


Performance evaluation of Component Based Software

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Companies developing high-tech products need to market products quickly and cost effectively, without compromising safety, performance and reliability.



The Allegio-project works closely with Philips Healthcare to evaluate and develop architecting methodologies for interventional X-ray (iXR) machines.

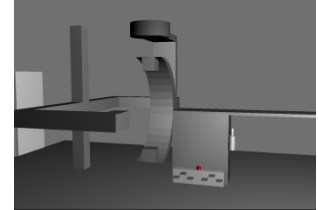
Research question: How to apply performance evaluation at an early-stage to reduce the number of mistakes made and gain confidence in product quality?

iXR machines make pictures from varying angles and at high frequencies, using X-ray technology. Motion and imaging constitute the machines.

Motion

Using a 3D-model, motion keeps track of the machine parts and the patient on the machine for collision prevention. Distance queries are used to continuously keep track of distances between objects and to keep them apart. Distance queries are computationally demanding. Hence, good performance is vital.

3D-model
of objects



Imaging

By applying sophisticated algorithms on the incoming X-ray signal, high quality images of a patient are generated. A doctor requires images to be displayed on the screen very soon after acquisition. Therefore, image processing needs to meet stringent performance requirements.

Imaging
screen

